

Assignment Guidelines for Applied Physics (SE-358)

Course Title: Applied Physics
Course Number: SE-358
Department of Computer Science
University of Karachi

Assignment Topics:

Section 1: Electromagnetism and Optics

- The Biot-Savart Law
- Ampère's Law
- Solenoids and Toroids
- Faraday's Experiments and Faraday's Law of Induction
- Lenz's Law
- Motional EMF (Electromotive Force)
- Induced Electric Fields
- The Basic Equation of Electromagnetism
- Induced Magnetic Fields
- Displacement Current
- Reflection and Refraction of Light Waves
- Total Internal Reflection
- Two-Source Interference
- Double-Slit Interference
- Interference from Thin Films
- Diffraction and the Wave Theory
- Single-Slit Diffraction
- Polarization of Electromagnetic Waves
- Polarizing Sheets

Note:

1. You are free to select one or more topics from the list above.
2. You may also choose an alternative topic, provided it is relevant to the subject matter.

Length: Section 1 should not exceed 1500 words (approximately 3-4 pages).

Section 2: Diodes and Circuit Simulation

- Name and discuss a circuit that uses a diode as its primary component.
- Explain its functions and working.

Bonus points: Simulate the circuit using Multisim, and include screenshots of the circuit as well as its respective waveforms. (Optional)

Length: Section 2 should not exceed 800 words (approximately 2 pages).

General guidelines:

1. Provide relevant citations or references in your work. You may use any standard referencing style (APA, MLA, or others).
2. Include relevant figures, diagrams, or tables where necessary. Ensure all visuals are properly numbered and labeled with captions.
3. Each group must submit one assignment.
4. Ensure the assignment is properly formatted and proofread.
5. Relevant headers and footers may be included.
6. Your title page should include the following information:
 - Relevant course information
 - Name of Department and University
 - Assignment Topic
 - Group Number
 - Names of all students in the group
 - Date of submission

MARKS DISTRIBUTION:

Theory (80)

- Assignment: 15 marks
- Terminal Exam: 65 marks

Lab (20)

- Lab Report: 05 marks
- Lab Quiz: 15 marks